

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R

PRACTICAL 15

AIM: 15. Generating basic summaries using str() or summary() (R).

PURVI ROSHAN KARKERA S086

PURVI ROSHAN KARKERA

S086

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
R452 - RPROG
> # =====
> # R Scripts: Generating Basic Summaries
> # Functions: str() and summary()
> # Dataset: arxiv_ai.csv
> # =====
> # 1. SETUP: Import Dataset
> # =====
> df <- read.csv("arxiv_ai.csv", na.strings = c("", "NA"))
> print("---- Data Loaded from arxiv_ai.csv ----")
[1] "---- Data Loaded from arxiv_ai.csv ----"
> head(df)
  authors
1 [arxiv.Result.Author("M. L. Ginsberg")]
2 [arxiv.Result.Author("M. P. Wellman")]
3 [arxiv.Result.Author("I. P. Gent"), arxiv.Result.Author("T. Walsh")]
4 [arxiv.Result.Author("F. Bergadano"), arxiv.Result.Author("D. Gunetti"), arxiv.Result.Author("U. Trinchero")]
5 [arxiv.Result.Author("J. C. Schlimmer"), arxiv.Result.Author("L. A. Henneke")]
6 [arxiv.Result.Author("M. Buchheit"), arxiv.Result.Author("F. W. Domin", arxiv.Result.Author("A. Schaeff")]
  comment doi
1 ["cs.AI"] See http://www.jair.org/ for an online appendix and other files\n accompanying this article <NA>
2 ["cs.AI"] See http://www.jair.org/ for any accompanying files <NA>
3 ["cs.AI"] See http://www.jair.org/ for any accompanying files <NA>
4 ["cs.AI"] See http://www.jair.org/ for any accompanying files <NA>
5 ["cs.AI"] See http://www.jair.org/ for an online appendix and other files\n accompanying this article <NA>
6 ["cs.AI"] See http://www.jair.org/ for any accompanying files <NA>
  entry_id journal_ref
1 http://arxiv.org/abs/cs/9308102v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 25-46
2 http://arxiv.org/abs/cs/9308102v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 1-23
3 http://arxiv.org/abs/cs/9309101v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 47-59
4 http://arxiv.org/abs/cs/9311101v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 91-107
5 http://arxiv.org/abs/cs/9311102v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 61-69
6 http://arxiv.org/abs/cs/9312101v1 Journal of Artificial Intelligence Research, Vol 1, (1993), 109-138
  pdf_url primary_category published
1 http://arxiv.org/pdf/cs/9308102v1 cs.AI 1993-08-01 00:00:00+00:00
2 http://arxiv.org/pdf/cs/9308102v1 cs.AI 1993-08-01 00:00:00+00:00
3 http://arxiv.org/pdf/cs/9309101v1 cs.AI 1993-09-01 00:00:00+00:00
4 http://arxiv.org/pdf/cs/9311101v1 cs.AI 1993-11-01 00:00:00+00:00
5 http://arxiv.org/pdf/cs/9311102v1 cs.AI 1993-11-01 00:00:00+00:00
6 http://arxiv.org/pdf/cs/9312101v1 cs.AI 1993-12-01 00:00:00+00:00
summary
1
Because of their occasional need to return to shallow points in a search tree, existing backtracking methods can sometimes erase meaningful progress toward solving a search problem. In this paper, we present a method by which backtracking points can be moved deeper in the search space, thereby avoiding this difficulty. The technique developed is a
```

Environment History Connections Git Tutorial

Global Environment

Object	Size
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

Values

Variable	Value
avg_comment_length	73.618481095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
total_records	10000L

Files Plots Packages Help Viewer Presentation

Home RPROG

Name	Size	Modified
gitignore	44 B	Dec 1, 2025, 11:03 AM
data	863 KB	Nov 24, 2025, 12:31 PM
history	457 B	Dec 1, 2025, 11:12 AM
AIImpact_on_Jobs_2030.csv	2964 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac 6.7.8.9.10		
prac 6.7.8.9&10 r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
arxiv_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

USD/INR +0.25% 11:41 08-12-2025

PURVI ROSHAN KARKERA

S086

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Source

```
R - R4.5.2 --RPROG--  
5 http://arxiv.org/pdf/cs/9311102v1 cs.AI 1993-11-01 00:00:00+00:00  
6 http://arxiv.org/pdf/cs/9312101v1 cs.AI 1993-12-01 00:00:00+00:00
```

summary
1
Because of their occasional need to return to shallow points in a search tree, existing backtracking methods can sometimes erase meaningful progress toward solving a search problem. In this paper, we present a method by which backtracking points can be moved deeper in the search space, thereby avoiding this difficulty. The technique developed is a variant of dependency-directed backtracking that uses only polynomial space while still providing useful control in formation and retaining the completeness guarantees provided by earlier approaches.
2
Market price systems constitute a well-understood class of mechanisms that under certain conditions provide effective decentralization of decision making with minimal communication overhead. In a market-oriented programming approach to distributed problem solving, we derive the activities and resource allocations for a set of computational agents by computing the competitive equilibrium of an artificial economy. WALRAS provides basic constructs for defining computational market structures, and protocols for deriving their corresponding price equilibria. In a particular realization of this approach for a form of multicommodity flow problem, we see that careful construction of the decision process according to economic principles can lead to efficient distributed resource allocation, and that the behavior of the system can be meaningfully analyzed in economic terms.
3
We describe an extensive study of search in GSAT, an approximation procedure for propositional satisfiability. GSAT performs greedy hill-climbing on the number of satisfied clauses in a truth assignment. Our experiments provide a more complete picture of GSAT's search than previous accounts. We describe in detail the two phases of search: rapid hill-climbing followed by a long plateau search. We demonstrate that when applied to randomly generated 3SAT problems, there is a very simple scaling with problem size for both the mean number of unsatisfied clauses and the mean branching rate. Our results allow us to make detailed numerical conjectures about the length of the hill-climbing phase, the average gradient of this phase, and to conjecture that both the average score and average branching rate decay exponentially during plateau search. We end by showing how these results can be used to direct future theoretical analysis. This work provides a case study of how computer experiments can be used to improve understanding of the theoretical properties of algorithms.
4
As real logic programmers normally use cut (!), an effective learning procedure for logic programs should be able to deal with it. Because the cut predicate has only a procedural meaning, clauses containing cut cannot be learned using an extensional evaluation method, as is done in most learning systems. On the other hand, searching a space of possible programs (instead of a space of independent clauses) is unfeasible. An alternative solution is to generate first a candidate base program which covers the positive examples, and then make it consistent by inserting cut where appropriate. The problem of learning programs with cut has not been investigated before and this seems to be a natural and reasonable approach. We generalize this scheme and investigate the difficulties that arise. Some of the major shortcomings are actually caused, in general, by the need for intensional evaluation. As a conclusion, the analysis of this paper suggests, on precise and technical grounds, that learning cut is difficult, and current induction techniques should probably be restricted to purely declarative logic languages.
5
To support the goal of allowing users to record and retrieve information, this paper describes an interactive note-taking system for pen-based computers with two distinctive features. First, it actively predicts what the user is going to write. Second, it automatically constructs a custom, button-box user interface on request. The system is an example of a learning-apprentice-software agent. A machine learning component characterizes the syntax and semantics of the user's information. A performance system uses this learned information to generate completion strings and construct a user interface. Description of Online Appendix: People like to record information. Doing this on paper is initially efficient, but lacks flexibility. Recording information on a computer is less efficient but more powerful. In our new note taking software, the user records information directly on a computer. Behind the interface, an agent acts for the user. To help, it provides defaults and constructs a custom user interface. The demonstration is a

Environment History Connections Git Tutorial

Global Environment

Object	Size
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

Values

Variable	Value
avg_comment_length	73.618461095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
total_records	10000L

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

Home RPROG

Name	Size	Modified
gitignore	44 B	Dec 1, 2025, 11:03 AM
README	863 KB	Nov 24, 2025, 12:31 PM
history	457 B	Dec 1, 2025, 11:12 AM
AI_impact_on_jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac 6.7.8.9.10		
prac 6.7.8.9&10 r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
ai_cv	13.3 MB	Dec 8, 2025, 10:58 AM

Type here to search

USD/INR +0.25% 11:42 08-12-2025

PURVI ROSHAN KARKERA

S086

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
R452 - ~/RPROG/
ral and reasonable approach. We generalize this scheme and investigate the difficulties that arise. Some of the major shortcomings are actually increased, in general, by the need for intentional evaluation. As a conclusion, the analysis of this paper suggests, on precise and technical grounds, that learning cut is difficult, and current induction techniques should probably be restricted to purely declarative logic languages.
5 To support the goal of allowing users to record and retrieve information, this paper describes an interactive note-taking system for pen-based computers with two distinctive features. First, it actively predicts what the user is going to write. Second, it automatically constructs a custom, button-box user interface on request. The system is an example of a learning-apprentice software agent. A machine learning component characterizes the syntax and semantics of the user's information. A performance system uses this learned information to generate completion strings and construct a user interface.
Description of Online Appendix: People like to record information. Doing this, however, is initially inefficient, but lacks flexibility. Recording information on a computer is less efficient but more powerful. In our new note-taking software, the user records information directly on a computer. Behind the interface, an agent acts for the user. To help, it provides defaults and constructs a custom user interface. The demonstration is a QuickTime movie of the note taking agent in action. The file is a binhexed self-extracting archive. Macintosh utilities for binhex are available from vmac.archive.unich.edu. QuickTime is available from ftp.apple.com in the /mactools/QuickTime/ directory.
6
Terminological knowledge representation systems (TKRSs) are tools for redesigning and using knowledge bases that make use of terminological languages (or concept languages). We analyze from a theoretical point of view a TKRS whose capabilities go beyond the ones of presently available TKRSs. The new features studied, often required in practical applications, can be summarized in three main points. First, we consider a highly expressive terminological language, called ALNOR, including general complements of concepts, number restrictions and role conjunction. Second, we allow to express inclusion statements between general concepts, and terminological cycles as a particular case. Third, we prove the decidability of a number of desirable TKRS deduction services (like satisfiability, subsumption and instance checking) through a sound, complete and terminating calculus for reasoning in ALNOR knowledge bases. Our calculus extends the general technique of constraint systems. As a byproduct of the proof, we get also the result that inclusion statements in ALNOR can be simulated by terminological cycles, if descriptive semantics is adopted.
1
2 title
3 Dynamic Backtracking
4 A Market-Oriented Programming Environment and its Application to Distributed Multicommodity Flow Problems
5 An Empirical Analysis of Search in GSAT
6 The Difficulties of Learning Logic Programs with Cut
7 Software Agents: Completing Patterns and Constructing User Interfaces
8 Decidable Reasoning in Terminological Knowledge Representation Systems
9
10 updated
11 1 1993-08-01 00:00:00+00:00
12 2 1993-08-01 00:00:00+00:00
13 3 1993-08-01 00:00:00+00:00
14 4 1993-11-01 00:00:00+00:00
15 5 1993-11-01 00:00:00+00:00
16 6 1993-12-01 00:00:00+00:00
17
18 >
19 > # =====
20 > # 2. USING str() (Structure)
21 > # =====
22 >
23 > # Purpose:
24 > # - Shows internal structure of dataset
25 > # - Data types of columns
26
```

Environment History Connections Git Tutorial

Global Environment

Object	Size
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

Values

Variable	Value
avg_comment_length	73.618461095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
total_records	10000L

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

Name	Size	Modified
gitignore	44 B	Dec 1, 2025, 11:03 AM
RData	86.3 KB	Nov 24, 2025, 12:31 PM
Rhistory	457 B	Dec 1, 2025, 11:12 AM
ALImpact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R	0 B	Dec 1, 2025, 12:21 PM
prac.6.7.8.9.10	0 B	Dec 1, 2025, 12:21 PM
prac.6.7.8.9&10.r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac.pdfs	0 B	Dec 1, 2025, 12:21 PM
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
anhr_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

Type here to search

USD/INR +0.25% ENG 11:42 08-12-2025

PURVI ROSHAN KARKERA

S086

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
R - R4.5.2 - ~/RPROG/
> # 2. USING str() (Structure)
> # Purpose:
> # - Shows internal structure of dataset
> # - Data types of columns
> # - Observations and variables
>
> print("---- OUTPUT OF str() ----")
[1] "---- OUTPUT OF str() ----"
> str(df)
'data.frame': 10000 obs. of 12 variables:
 $ authors      : chr  "[arxiv.Result.Author('M. L. Ginsberg')]" "[arxiv.Result.Author('M. P. Wellman')]" "[arxiv.Result.Author('I. P. Gent')]" "[arxiv.Result.Author('T. Walsh')]" "[arxiv.Result.Author('F. Bergadano')]" "[arxiv.Result.Author('D. Guretti')]" "[arxiv.Result.Author('U. Trinchero')]" ...
 $ categories    : chr  "['cs.AI']" "['cs.AI']" "['cs.AI']" ...
 $ comment       : chr  "See http://www.jair.org/ for an online appendix and other files\n accompanying this article" "See http://www.jair.org/ for any accompanying files" ...
 $ doi           : chr  "NA NA NA NA ..."
 $ entry_id      : chr  "http://arxiv.org/abs/cs/9308101v1" "http://arxiv.org/abs/cs/9308102v1" "http://arxiv.org/abs/cs/9308101v1" "http://arxiv.org/abs/cs/9311101v1" ...
 $ journal_ref   : chr  "Journal of Artificial Intelligence Research, Vol 1, (1993), 25-46" "Journal of Artificial Intelligence Research, Vol 1, (1993), 1-23" "Journal of Artificial Intelligence Research, Vol 1, (1993), 91-107" ...
 $ pdf_url      : chr  "http://arxiv.org/pdf/cs/9308101v1" "http://arxiv.org/pdf/cs/9308102v1" "http://arxiv.org/pdf/cs/9308101v1" "http://arxiv.org/pdf/cs/9311101v1" ...
 $ primary_category : chr  "cs.AI" "cs.AI" "cs.AI" "cs.AI" ...
 $ published     : chr  "1993-08-01 00:00:00+00:00" "1993-08-01 00:00:00+00:00" "1993-09-01 00:00:00+00:00" "1993-11-01 00:00:00+00:00" ...
 $ summary      : chr  "Because of their occasional need to return to shallow points in a search tree, existing backtracking methods [...] truncated. Market price systems constitute a well-understood class of mechanisms that under certain conditions provide efficient [...] truncated. We describe an extensive study of search in GSAT, an approximation procedure for propositional satisfiability. [...] truncated. As real logic programmers normally use cut (1), an effective learning procedure for logic programs should be a [...] truncated. ...
 $ title         : chr  "Dynamic Backtracking: A Market-Oriented Programming Environment and its Application to Distributed Multimodality Flow Problems" "An Empirical Analysis of Search in GSAT" "The Difficulties of Learning Logic Programs with Cut" ...
 $ updated      : chr  "1993-08-01 00:00:00+00:00" "1993-08-01 00:00:00+00:00" "1993-09-01 00:00:00+00:00" "1993-11-01 00:00:00+00:00" ...
>
> # 3. USING summary() (Statistical Summary)
>
> print("---- OUTPUT OF summary() [Before Factor Conversion] ----")
[1] "---- OUTPUT OF summary() [Before Factor Conversion] ----"
```

Environment

Object	Class	Attributes
job_title	factor	3000 obs. of 18 variables
long_df	data.frame	20000 obs. of 4 variables
processed_data	data.frame	10000 obs. of 22 variables
unique_titles	data.frame	5 obs. of 3 variables
wide_df	data.frame	10000 obs. of 4 variables

Values

Variable	Value
avg_comment_length	73.618461095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
total_records	10000L

Files

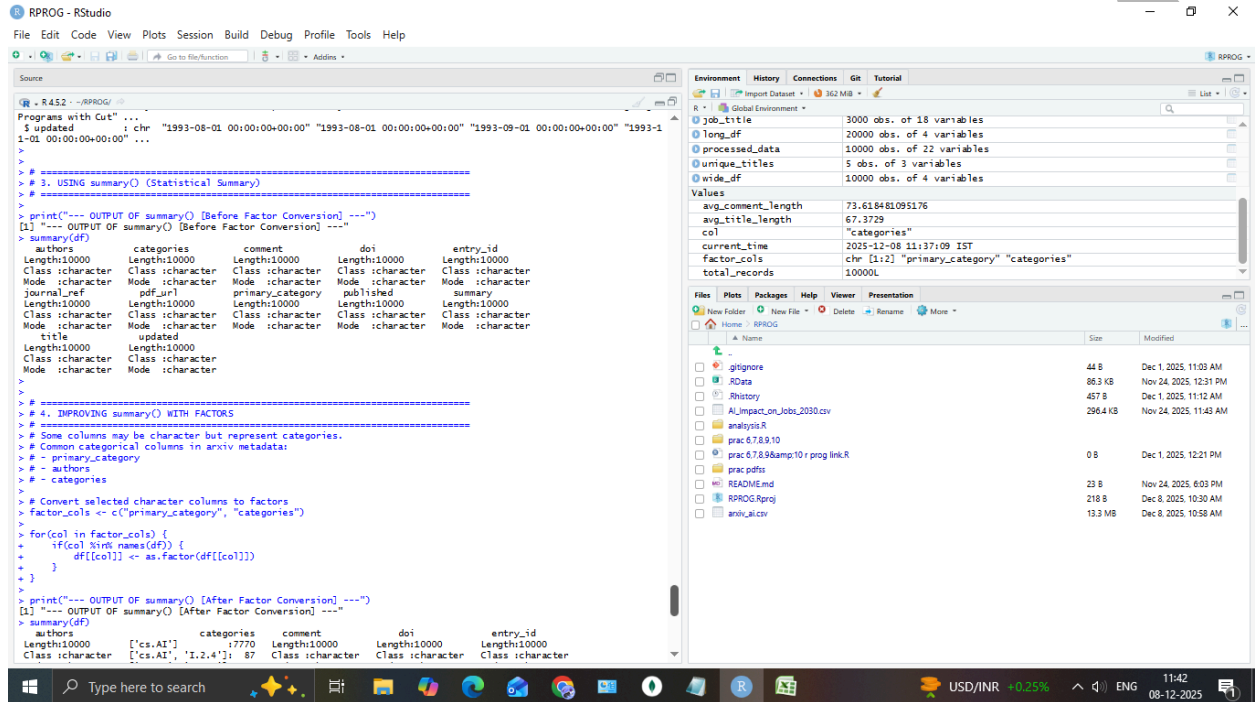
Name	Size	Modified
gitignore	44 B	Dec 1, 2025, 11:03 AM
RData	86.3 KB	Nov 24, 2025, 12:31 PM
Rhistory	457 B	Dec 1, 2025, 11:12 AM
ALImpact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac 6.7.8.9.10		
prac 6.7.8.9&10 r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
arxiv_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

Taskbar: Type here to search, USD/INR +0.25%, 11:42, 08-12-2025

PURVI ROSHAN KARKERA

S086

SUBJECT NAME:Data Analysis with SAS / SPSS /R



SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R 4.5.2 - ~/RPROG/
> factor_cols <- c("primary_category", "categories")
> for(col in factor_cols) {
+   if(col %in% names(df)) {
+     df[[col]] <- as.factor(df[[col]])
+   }
+ }
> print("---- OUTPUT OF summary() [After Factor Conversion] ----")
[1] "---- OUTPUT OF summary() [After Factor Conversion] ----"
> summary(df)
      authors      categories      comment      doi      entry_id
Length:10000      Length:10000      Length:10000      Length:10000      Length:10000
Class: character      Class: character      Class: character      Class: character      Class: character
Mode: character      Mode: character      Mode: character      Mode: character      Mode: character

      journal_ref      pdf_url      primary_category      published      summary      title
Length:10000      Length:10000      Length:10000      Length:10000      Length:10000      Length:10000
Class: character      Class: character      Class: character      Class: character      Class: character      Class: character
Mode: character      Mode: character      Mode: character      Mode: character      Mode: character      Mode: character

      updated
Length:10000
Class: character
Mode: character

      (Other) : 1866
      cs.AI : 19280
      cs.LO : 104
      cs.NE : 104
      cs.LG : 86
      cs.CL : 52
      cs.CV : 52
      (Other) : 322

> #####
> # 5. Accessing Specific Summaries
> #####
> # Example numerical summaries (if columns exist)
> if("title" %in% names(df)) {
+   avg_title_length <- mean(nchar(df$title), na.rm = TRUE)
+   print(paste("Average Title Length:", avg_title_length))
+ }
[1] "Average Title Length: 67.3729"
> if("published" %in% names(df)) {
```

PURVI ROSHAN KARKERA

S086

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R



RPROG - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
l'cs.AI', 'I.2.4'): 87 Class icharacter Class icharacter Class icharacter
['cs.NE', 'cs.AI']: 75 Mode icharacter Mode icharacter Mode icharacter
['cs.AI', 'cs.LO']: 67
['cs.LO', 'cs.AI']: 63
['cs.AI', 'I.2.3']: 52
(Other): 11866
pdfurl primary_category published summary title
Length10000 Length10000 Length10000 Length10000 Length10000
Class icharacter Class icharacter Class icharacter Class icharacter Class icharacter
Mode icharacter Mode icharacter Mode icharacter Mode icharacter Mode icharacter
cs.AI : 9280
cs.LO : 104
cs.NE : 104
cs.LG : 86
cs.CL : 52
cs.CV : 52
(Other): 322

updated
Length10000
Class icharacter
Mode icharacter
```

Environment

Object	Variables
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

Values

Variable	Value
avg_comment_length	73.618481095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
total_records	10000L

Files

Name	Size	Modified
gitignore	44 B	Dec 1, 2025, 11:03 AM
RData	86.3 KB	Nov 24, 2025, 12:31 PM
Rhistory	457 B	Dec 1, 2025, 11:12 AM
ALImpact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
analysis.R		
prac.6.7.8.9.10		
prac.6.7.8.9&10.r prog link.R	0 B	Dec 1, 2025, 12:21 PM
prac.pdfs		
README.md	23 B	Nov 24, 2025, 6:03 PM
RPROG.Rproj	218 B	Dec 8, 2025, 10:30 AM
anhi_ai.csv	13.3 MB	Dec 8, 2025, 10:58 AM

```
> #####
> # 5. Accessing Specific Summaries
> #####
> # Example numerical summaries (if columns exist)
> if("title" %in% names(df)) {
+   avg_title_length <- mean(nchar(df$title), na.rm = TRUE)
+   print(paste("Average Title Length:", avg_title_length))
+ }
[1] "Average Title Length: 67.3729"
> if("published" %in% names(df)) {
+   total_records <- nrow(df)
+   print(paste("Total Records:", total_records))
+ }
[1] "Total Records: 10000"
> # If there is a "comment" column (character)
> if("comment" %in% names(df)) {
+   avg_comment_length <- mean(nchar(df$comment), na.rm = TRUE)
+   print(paste("Average Comment Length:", avg_comment_length))
+ }
[1] "Average Comment Length: 73.618481095176"
```

USD/INR +0.25% 11:42 08-12-2025

PURVI ROSHAN KARKERA

S086